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Founder & Full-Stack Developer Master Operating Manual

End-to-End Technical Guide: Architecture, Multi-Tenant Database, Cloud Automation, UPI AutoPay, and 24/7 Google Review AI

1. System Vision: The Solo Founder & Developer Blueprint

As the solo founder and developer of **Zentap.ai (Digital Solutions)**, your goal is to build a system that runs **100% autonomously in the cloud**. You cannot spend your day manually editing links, creating QR codes, chasing payments, or manually writing Google review replies. Zentap is architected as a hands-off, multi-tenant cloud platform where businesses sign up, get custom standees, have their subscriptions auto-debited, and have their Google Business Profile reviews automatically monitored and answered by AI 24 hours a day, 7 days a week.

> ***Core Zentap Value Proposition & Market Message:***
> *"Zentap helps local businesses get more Google reviews, starting with a free 14-day pilot."*
> • ***Hardware:*** *Custom A6 Acrylic Standee + NTAG213 NFC chip.*
> • ***Software Retainer:*** *■2,999/month recurring UPI AutoPay for review automation + private negative feedback gating.*

2. The Solo-Developer Tech Stack (What Software & Services to Use)

To eliminate server maintenance, use modern serverless edge architecture. Every tool below has been chosen specifically because its free tier covers your first 50 to 100 paying customers without costing you a single rupee in server rent.

Layer	Software / Tool	Role in Zentap Platform	Developer Setup
Frontend & API	**Next.js 14 (App Router)**	Hosts landing page, dynamic customer review URLs (/r/:slug), and merchant portal	TypeScript, React, Tailwind CSS
Cloud Hosting	**Vercel** (Edge Network)	Instant CI/CD from GitHub. Edge redirects serve standee taps in <150ms globally	Free tier (100GB bandwidth)
Database & Auth	**Supabase** (PostgreSQL)	Multi-tenant database storing merchant configs, scan analytics, review logs, and auth	Free tier (500MB DB, generous)
Payments (India)	**Razorpay Subscriptions**	Automated recurring monthly ■2,999 debits via UPI AutoPay (PhonePe, GPay, Paytm)	Standard 2% transaction fee
Review Engine	**Google Gemini 1.5 Flash**	Instant AI review draft generation in Gujarati/Hindi/English + autonomous owner replies	~■50 per 10,000 calls
Google Integration	**Google Business Profile API**	OAuth connection to client GMB account + Pub/Sub real-time review webhook listener	Google Cloud Console (Free)
WhatsApp Alerts	**AiSensy / Wati API**	Sends immediate WhatsApp ping to store owner whenever <4★ negative feedback is received	Official Meta WhatsApp API
Hardware Print	**@react-pdf/renderer**	Generates printable A6 standee PDF inserts with live QR automatically on merchant signup	Node serverless function

3. Multi-Tenant Database Architecture (Managing All Customers at Once)

To manage 50, 100, or 1,000 stores from a single admin panel without mixing customer data, use a **single-database, multi-tenant schema** in Supabase. Every store has a unique merchant_id and URL slug (e.g. virat-gold).

Core PostgreSQL Tables in Supabase:

1. merchants Table: Stores business profiles, branding colors, Google Place IDs, and OAuth tokens.
id (UUID PK), slug (VARCHAR UNIQUE), business_name, category, phone, whatsapp, place_id, google_refresh_token, theme_json, services_json, created_at

2. subscriptions Table: Manages Razorpay recurring subscription state.
id (UUID PK), merchant_id (FK), razorpay_sub_id, razorpay_customer_id, plan_id, status ('trial'|'active'|'past_due'|'cancelled'), trial_ends_at, next_billing_at

3. scan_events Table: Logs every physical NFC tap and QR scan.
id (UUID PK), merchant_id (FK), scan_type ('nfc'|'qr'), user_agent, ip_hash, rating_given, review_copied (BOOL), created_at

4. private_feedback Table: Catches ratings <4★ privately to shield Google rating.
id (UUID PK), merchant_id (FK), rating, customer_name, customer_phone, issue_text, status ('pending'|'contacted'|'resolved'), created_at

5. gmb_reviews_and_replies Table: Tracks Google reviews and autonomous AI responses.
id (UUID PK), merchant_id (FK), gmb_review_id, reviewer_name, star_rating, comment, ai_reply_text, reply_posted_at, status

4. Automated Subscription Billing Engine (UPI AutoPay via Razorpay)

In India, credit cards fail on recurring auto-debit due to OTP mandates. Zentap leverages **Razorpay Subscriptions with UPI AutoPay**. Once a client authorizes the mandate on PhonePe or Google Pay, monthly billing is 100% automated.

The Automated 14-Day Pilot to Monthly Retainer Lifecycle:

Step 1: Merchant Mandate Setup (Onboarding)
During signup on zentap.in, your Next.js API calls razorpay.subscriptions.create() with a 14-day free trial period. The store owner approves a ■1 pre-authorization on PhonePe or GPay. Days 1 to 14 are billed at ■0.

Step 2: Automated First Debit (Day 15)
At 12:01 AM on Day 15, Razorpay automatically debits ■2,999 from the client's bank account. Razorpay sends an HTTP webhook subscription.charged to your Vercel endpoint /api/webhooks/razorpay.

Step 3: Webhook Verification & State Update
Your webhook endpoint verifies the cryptographic signature (X-Razorpay-Signature). Upon verification, it updates Supabase: sets status = 'active' and increments next_billing_at by 30 days.

Step 4: Dunning & Failed Payment Recovery
If the client's account has insufficient funds, Razorpay fires payment.failed. Zentap automatically triggers a WhatsApp alert: *'Your Zentap standee service payment was unsuccessful. Click here to update your UPI ID and avoid standee deactivation.'* If unpaid after 72 hours, the redirect link zentap.in/r/:slug temporarily shows a service renewal prompt.

5. The 24/7 Auto Google Review Detection & AI Auto-Reply Pipeline

Google rewards active listings that reply to 100% of reviews with higher local Google Maps ranking. Here is the exact technical pipeline to detect and answer reviews hands-free using the official Google Business Profile API:

Architecture of the Auto-Reply Pipeline:

1. Merchant OAuth Authorization:
Store owner clicks *'Connect Google Business Profile'* in their Zentap dashboard. Next.js initiates Google OAuth 2.0 with scopes:
https://www.googleapis.com/auth/business.manage. The resulting refresh_token is AES-256 encrypted and stored in Supabase.

2. Google Cloud Pub/Sub Webhook Listener:
In Google Cloud Console, subscribe to GMB review notifications. Whenever a customer posts a review on Google Maps, Google pushes a message to your Vercel endpoint: /api/webhooks/gmb-review.

3. AI Sentiment Analysis & Response Generation:
The webhook extracts review stars, customer name, and text. Next.js calls Gemini 1.5 Flash with prompt:
'You are the store manager of [Business Name] in [City]. A customer left this [Star Rating] review: [Review Text]. Write a warm, professional, 40-word response. Mention 22K gold purity / bridal jewellery / fair pricing to boost local SEO.'

4. Publishing Reply Directly to Google Maps:
Using the merchant's stored refresh token, Zentap gets a fresh access token and calls:
PUT https://mybusiness.googleapis.com/v4/accounts/{accountId}/locations/{locationId}/reviews/{reviewId}/reply
Payload: { "comment": "Dear Valued Customer, thank you for your kind words! ..." }
The response is publicly live on Google Maps within 15 seconds of the customer posting.

6. Automated Standee Generation & Print-on-Demand Dropshipping

You do not need to assemble standees at home. When a merchant signs up, automate the physical supply chain:

- **Automated PDF Generation:** Serverless function renders a 300 DPI A6 artwork containing the store logo, gold border styling, and scannable QR code linking to zentap.in/r/:slug.
- **Dropship Printing Partner:** Establish an account with a local acrylic printer in Ahmedabad or Mumbai (sourced via IndiaMART). Wholesaler cost: ■70–■110 per A6 standee + NTAG213 sticker.

• **1-Click Order Webhook:** On subscription creation, Zentap emails the generated PDF and client delivery address directly to the printer. Vendor prints, sticks the programmed NFC tag, and dispatches via Delhivery or Shiprocket.

7. Step-by-Step Developer Implementation Roadmap

Follow this exact sprint-by-sprint development roadmap to bring the entire Zentap cloud system live:

Sprint	Target Milestone	Exact Developer Tasks	Success Output
Sprint 1 (Days 1–3)	**Pilot Deployment & Anchor Client**	Deliver the physical A6 acrylic standee to **Virat Gold Palace**. Test live QR (#lrd URL) and Gujarati AI draft assistant at their counter.	First live standee active on counter collecting reviews
Sprint 2 (Days 4–7)	**Cloud Stack & Database Setup**	1. Initialize Next.js 14 repo on GitHub. 2. Create Supabase project & execute SQL schema. 3. Point zentap.in domain to Vercel. 4. Build dynamic route /r/[slug].	zentap.in live on edge network with sub-150ms redirects
Sprint 3 (Days 8–11)	**Subscription & UPI AutoPay Engine**	1. Create ■2,999/mo plan on Razorpay Dashboard. 2. Build checkout page with 14-day free trial mandate. 3. Implement /api/webhooks/razorpay to toggle merchant status.	Automated recurring monthly debits running without manual invoicing
Sprint 4 (Days 12–16)	**Google API & AI Auto-Reply**	1. Configure Google Cloud Console OAuth 2.0. 2. Connect Pub/Sub webhook for new reviews. 3. Wire Gemini Flash prompt & API PUT reply. 4. Connect AiSensy for negative review WhatsApp alerts.	100% hands-free Google review replies within 15 seconds
Sprint 5 (Days 17–30)	**Retail Outreach & Scaling to 10 Stores**	Use the Virat Gold Palace results as your sales anchor. Pitch nearby salons, clinics, and cafes on Nikol Gam Road and CG Road. Offer 14-day free pilot.	**10 Stores Active = ■29,990 / month MRR**

Founder & Developer Final Guidance:
*Do not wait until the entire cloud software is 100% finished before talking to merchants. The physical standee and the working demo page we built for **Virat Gold Palace** are already complete and functional. Place the physical standee on their billing counter tomorrow. Once the owner sees 15–20 new 5-star Google reviews rolling in, closing them on the ■2,999/month recurring plan becomes effortless. Build the automation behind you as the cash flow begins.*